

Raaghav Thirumaligai

Mechanical Engineer

✉ rthirumaligai@gmail.com | ✉ raaghavt@icloud.com | [in](#) LinkedIn | 📞 (818) 651-2193

OBJECTIVE

Controls-focused mechanical engineer (PhD '30) with hands-on mechatronics and vehicle systems experience (FSAE EV). Built real-time control and estimation projects in MATLAB/Simulink (inverted pendulum: swing-up + LQR/LQG) and developed instrumentation + automation workflows (Arduino-MATLAB DAQ, Python + vision-based detection).

Seeking controls / automation internship roles in the Bay Area (model-based design, control theory, real-time).

SKILLS

Control Theory – State-space Linear Systems, Lyapunov stability, LQR/LQG, MIMO analysis
Nonlinear Control – Stability analysis via class K functions; feedback linearization, input-output ideas
Model Predictive Control – Constraint handling, finite-horizon optimization, trajectory optimization
Estimation / Filtering – Observers, Kalman filtering (LQG); sensor calibration, DAQ, filtering for experiments
Decision / Game Theory – Zero-sum / bimatrix games, saddle-point concepts; Minimax Theorem; dynamic games
Robust Control – System norms, small-gain theorem, H^∞ synthesis, mixed sensitivity
Hybrid Systems – Continuous-discrete modeling, hybrid analysis & design
Programming – MATLAB/Simulink (System ID and Control); Python (ML / sensors); Arduino; Git
CAD / Analysis – SolidWorks & Fusion 360; GD&T; assemblies; drawings; FEA
Fabrication – Rapid prototyping, fixtures, waterjet/laser, sheet metal, 3D printing (FDM/SLA)

EDUCATION

- **University of California, Santa Barbara - Ph.D. Mechanical Engineering** Projected Jun 2029
GPA 3.91/4.0 — Research Interest: Control Theory
- **University of California, Santa Barbara - M.S. Mechanical Engineering** Jul 2025
GPA 3.90/4.0 — Project: [Clogging of fiber suspensions at constrictions](#)
- **University of California, Santa Barbara - B.S. Mechanical Engineering** Jun 2025
GPA 3.76/4.0 — Senior Capstone: [600 V Vehicle Battery Container](#)

PROJECTS

Constrained Satellite Rendezvous With MPC (MATLAB/MPCTools) Jan 2026 - Mar 2026

- Built a Hill-Clohessy-Wiltshire (HCW) relative-motion model and derived a zero-order-hold discrete-time state-space system for receding-horizon control design.
- Implemented and benchmarked saturated LQR, state-focused constrained MPC, and economic MPC; evaluated convergence, feasibility behavior, and fuel proxy over random initial conditions.
- Presented controller tradeoffs (fast convergence vs fuel minimization) in final report/slides; project page: [website](#).

Inverted Pendulum on Cart: swing-up + balancing + catch (MATLAB/Simulink) Apr 2025 - Jun 2025

- Identified linear models around stable and unstable equilibria using logarithmic sine sweeps and MATLAB System Identification (tfest); validated via frequency response/root-locus checks.
- Implemented a state-machine controller: energy-based swing-up, LQR/LQG (Kalman filter) balancing near upright, and LQR/LQG "catching" to return to the downward equilibrium.

King Rook King Endgame: Noncooperative Game Theory Implementation (Matlab) Feb 2026 - Mar 2026

- Modeled king+rook vs king as a finite, sequential, zero-sum dynamic game from ECE 270 (Joao P. Hespanha).
- Implemented legal-move generation, terminal detection (checkmate/stalemate), depth-limited minimax, alpha-beta pruning, and persistent cache reuse.
- Designed payoff shaping with terminal rewards/penalties and black-king mobility (oxygen) heuristics.

RESEARCH EXPERIENCE

Biological Control Lab, Yeung UCSB - *Research Assistant*Jan 2025 - Jan 2026

- Co-developed a control-oriented model of genome-wide transcription coupled to DNA supercoiling dynamics regulated by topoisomerases.
- Supported analysis of stochastic and nonlinear population models (stability/sensitivity analysis) to identify parameters influencing transcription and growth dynamics.
- Co-author on manuscript accepted (Jan 2026); preprint: [10.1101/2025.05.26.655193](#).

M&M Flow Lab, Sauret UCSB - *Master's Project*Aug 2024 - Jun 2025

- Researching clogging dynamics in particle-laden suspensions in constricted flows, relevant to medical/industrial systems.
- Prototyped a vision-based clog detector using a rolling brightness/windowed-visibility metric; used it to support logging and post-analysis of clogging events.
- Presented findings at November 2024 APS Division of Fluid Dynamics meeting.

M&M Flow Lab, Sauret UCSB - *Undergraduate Researcher*Jan 2023 - Aug 2024

- Explored cohesive forces in granular materials with two PhD student mentors to help understand inter-particle mechanics from the introduction of cohesive forces.
- Designed and built an instrumented annular shear cell; developed an Arduino–MATLAB DAQ + calibration workflow for strain-gauge measurements and repeatable experiments.
- Presented work at Southern California Flow Physics Symposium.

Fluid Energy Science Lab, Luzzatto-Fegiz UCSB - *Undergraduate Researcher*Jun 2023 - Oct 2023

- 3D-printed and laser-cut annuli to simulate kites for wind energy applications.
- Data collection for wake model verification in a wind tunnel; made an Arduino-MATLAB DAQ system and app.

EXPERIENCE - WORK

Teaching Assistant - UCSB: Mech E Academic Student EmployeeApr 2024 - Present

- Grading and holding office hours for Vibrations course covering signals and dynamicsJan - Mar 2026
- Led Mechanical Engineering Laboratory Section covering experimental procedures and data analysisSept - Dec 2025
- Taught Dynamics Office Hours covering FBD and Energy based system analysisApr - Jun 2025
- Graded assignments and advised bio-inspired design groupsJan - Mar 2025
- Assisted in grading for Dynamics course and Fluids courseApr - Dec 2024

Resident Assistant - UCSB: San Rafael Residence HallAug 2022 - Jun 2023

- Managed resident support, conflict resolution, and event planning in a high-traffic dorm environment.

GameMaster - Diablo Escapes, Walnut CreekJul 2020 - Aug 2022

- Guided customers through VR escape rooms; troubleshoot hardware/software issues and explained unfamiliar tech.
- Helped design puzzles for physical escape rooms under construction.

HONORS

- Undergraduate Engineering Honors ProgramSept 2023 - Jun 2024
- Dean's Honors (ENGR) x5 per quarter
- Undergraduate Research Awards x2 (\$3750 and \$2500)2024 and 2025